

DATA MANAGEMENT PLAN

Template	Horizon Europe Data Management Plan v1.1 (01.04.2022)
Project acronym	CNM
Project name	Semiconductor Device Characterization Data (180 nm CMOS)
Deliverable	DMP (repository-level; standalone research dataset)
Version	1.2
Date	10/06/2025
Dissemination level	Public
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Data steward	uzzielperez25@gmail.com
Repository	https://github.com/uzzielperez/CNM

History of Changes

Version	Publication date	Change
1.0	10/06/2025	Initial DMP; FAIR inventory; EU access policy
1.1	10/06/2025	Analysis pipelines (src/); formal HE PDF export
1.2	10/06/2025	JSON experiment sidecars (metadata/); schema documented

1. Data Summary

1.1 Purpose of Data Generation and Relation to Project Objectives

The CNM repository collects and curates semiconductor device characterization data for 180 nm CMOS technology. The data support device-level and circuit-level research objectives:

- Extraction of DC parameters (threshold voltage, I_{on} , I_{off} , subthreshold swing)
- Validation of process-corner and Monte Carlo simulation setups
- Analysis of $1/f$ noise and AC frequency response
- Wafer-level yield and figures-of-merit benchmarking

Data enable comparison of measured versus simulated device behaviour, production of publication figures, and documentation of calibration conditions for reproducibility.

Technology context: 180 nm PDK; nominal supply 1.8 V; process corners tt, ff, ss, fs, sf.

1.2 Re-use of Existing Data

No third-party datasets are currently reused. External 180 nm PDK model files are referenced in documentation but not redistributed due to foundry intellectual property restrictions (see Sections 2.2 and 6).

Asset ID	Filename	Generated / Reused	Origin	Purpose
D1	march8.csv	Generated	Lab IV measurement	DC sweep validation
D2	copy of copy of IV_curves.xlsx.csv	Generated	Spreadsheet export	Raw IV reference
D3	noise_NEW2.dat	Generated	Noise analyser	$1/f$ noise characterisation
D4	asdfgraph.png.csv	Generated	Plot export	Trace archival
D5	data1.csv	Generated	Bench measurement	Sample V–I curve
D6	Measurements - Juan - version2 - corrected.csv	Generated	Post-processed lab data	Calibrated DC parameters
D7	untitled2.csv	Generated	Switch bench	Ron/Roff ratios
D8	FINAL_v3_USETHIS.csv	Generated	AC analyser + processing	Figure-ready frequency response
D9	definitive_DEFINITIVE.csv	Generated	Wafer test	Yield / f_{max} benchmarking
D10	docs/*.txt	Generated	Lab notes	Provenance and methods

1.3 Types and Formats of Data

Data type	Format	Generation software	Open format
Tabular measurements	CSV (UTF-8, comma-separated)	Instruments, spreadsheets, Python/pandas	Yes
Noise spectra	DAT (ASCII, whitespace-delimited)	Noise analyser export	Yes
Lab notes	TXT (UTF-8)	Plain text editor	Yes
Analysis code	PY, M, OGS	Python, MATLAB, Origin LabTalk	Yes

Preferred formats follow UK Data Service recommendations: plain text and non-proprietary tabular formats.

1.4 Provenance

Pipeline stage	Repository path	Rule
Raw instrument output	data/01_raw/	Read-only; append dated exports only
Calibrated / filtered	data/02_processed/	Document transformations in src/ or docs/
Figure-ready matrices	data/03_final/	Versioned; cite in publications
Methods and calibration	docs/	Linked from dataset metadata

Each dataset records who collected it, when, with which instrument, under which bias/temperature, and what processing was applied.

1.5 Expected Data Size

Category	Approximate size
Current repository	< 5 MB
Per measurement campaign	< 50 MB
Full project (estimated)	< 500 MB

1.6 Data Utility

Target users: semiconductor device researchers, analog/RF circuit designers, reliability and yield engineers, reproducibility reviewers, and EOSC metadata harvesters.

Keywords: semiconductor characterization, CMOS, 180nm, IV curves, 1/f noise, threshold voltage, wafer yield, process corners, Monte Carlo, FAIR data.

2. FAIR Data

2.1 Making Data Findable, Including Provisions for Metadata

Measure	Status	Planned action
Persistent identifier (DOI)	Planned	Zenodo deposition (EOSC-aligned)
Version control	Active	Git tags per release (e.g. v1.0.0)
Descriptive metadata	Active	README + this DMP
Sidecar metadata	Active	metadata/*.json per experiment (hardware, conditions, data_files)
Machine-readable catalogue	Active	metadata/index.json
Discovery indexing	Partial	GitHub; full after Zenodo DOI

JSON Experiment Sidecars (metadata/)

Each major dataset has a companion JSON file in metadata/ that acts as a human-readable, machine-parsed bridge between lab records and analysis pipelines (Python, MATLAB, Origin). This supports FAIR findability and programmatic ingestion without hard-coded file paths.

Schema (metadata_version 1.0.0):

Key	Content
metadata_version	Schema semver
experiment_id	Unique ID, e.g. 2025-CNM-IV-001
hardware_setup	Instruments, DUT, gain, technology node
experimental_conditions	Temperature, bias, corner, measurement date
data_files	Repo-relative paths to raw, processed, and final files

Conventions: lowercase snake_case keys; SI units in key names (_v, _hz, _c, _db); UTF-8 encoding. **Catalogue:** metadata/index.json. **Current sidecars:** march8_iv_sweep.json, noise_NEW2.json, final_ac_response.json, definitive_yield.json. **Full specification:** metadata/README.md.

Minimum metadata per Zenodo deposition: title, creators, description, keywords, licence (CC-BY-4.0), version, publication date, related identifiers. Dublin Core and DataCite-compatible fields will be used. JSON sidecars complement deposition records with per-experiment instrument and path detail. Metadata will be harvestable via Zenodo and GitHub.

2.2 Making Data Accessible

2.2.1 Repository

Primary working repository: GitHub (<https://github.com/uzzielperez/CNM>). Archival repository (planned): Zenodo — trusted, metadata-ready, EOSC-federated general-purpose repository.

2.2.2 Data

Principle: As open as possible, as closed as necessary (Horizon Europe Programme Guide).

Data class	Access	Licence	Restriction
Measurement CSV/DAT/TXT	Open	CC-BY-4.0	None; no personal data
Lab notes	Open	CC-BY-4.0	Redact third-party names on request

180 nm PDK / foundry models	Not distributed	N/A	IPR / contractual (foundry NDA)
Pre-publication embargo	None currently	—	Max. embargo if patents filed

Access protocol: HTTPS git clone and direct download. No authentication required for open datasets.

2.2.3 Metadata

Metadata openly available under CC0. Metadata includes access conditions and contact point even if datasets are later restricted. Contact: uzzielperez25@gmail.com

2.3 Making Data Interoperable

Repository structure:

- data/01_raw/ — factory and instrument outputs (read-only)
- data/02_processed/ — calibrated tables
- data/03_final/ — publication-ready matrices
- src/ — analysis pipelines (Python, MATLAB, Origin)
- DataManagementPlan/ — formal DMP (this document)
- docs/ — calibration sheets and lab notes
- metadata/ — JSON experiment sidecars (index.json + per-run records)

Interoperability: UTF-8 encoding; SI units in headers; consistent column naming (vgs, vds, ids, freq_hz, temp_c); CSV RFC 4180 compliance. JSON sidecars in metadata/ link tabular files to hardware and experimental context so pipelines can resolve paths programmatically (see Section 2.1).

2.4 Increase Data Re-use

Output	Licence	Citation requirement
Datasets	CC-BY-4.0	DOI + version tag
Software (src/)	MIT	Repository + release tag
Documentation	CC-BY-4.0	Same as datasets

Reuse tools: Python 3.10+ (pandas), MATLAB R2021a+, Origin 2021+. See repository README Section 9 and src/ for pipelines. Downstream users must cite the DOI, not redistribute PDK content, and document derivative processing.

3. Other Research Outputs

Output	Location	Preservation
Analysis scripts	src/	Git + Zenodo archive
Figures from 03_final/	Publications / Zenodo	Link to source CSV
SPIICE netlists	src/ or docs/	Git; exclude NDA decks
Presentations	DataManagementPlan/ or Zenodo	Supplementary material

Software documentation follows Lee et al. (2014) *Ten simple rules for documenting scientific software*.

4. Allocation of Resources

Activity	Responsible	Effort
Data curation	Data steward	~2 h/month
DMP / README updates	Data steward	Quarterly

Zenodo DOI registration	Data steward	~1 h per release
Backup verification	Data steward	Annually
Storage	GitHub + Zenodo free tier	€0 at current volume

5. Data Security

Risk	Mitigation
Overwrite of raw data	data/01_raw/ append-only; Git history
Data loss	GitHub remote + Zenodo snapshots
Unauthorised modification	Branch protection; immutable release tags
Confidential leakage	No PDK/foundry files in repository
Local workstation loss	Push after each curated batch

Standard institutional IT policies apply. No sensitive personal data stored (see Section 6).

6. Ethics

Topic	Assessment
Human subjects	Not applicable — device measurements only
GDPR	Not applicable for current assets; reassess if personal data added
Intellectual property	Measurement data CC-BY-4.0; PDK is third-party IP
Export control	Characterization results only; review if fabrication details added
Ethics committee	Not required for ex vivo device testing

7. Other Issues

- Legacy path datasets/ is being migrated to data/01_raw/, data/02_processed/, data/03_final/.
- Zenodo DOI and ORCID to be added before formal grant submission.
- Grant agreement number and institution to be inserted when CNM is linked to a funded Horizon Europe project.
- This DMP is a living document; review annually or before each Zenodo release.

End of Data Management Plan · CNM v1.1 · 10/06/2025 · Public